

docloom: LLM to Documents, Instantly

Validated IR, Deterministic Renderers, 6 Output Formats

docloom project — 2026-07-21

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Validated IR + Deterministic Renderers = 6 Native Formats

From Raw LLM Output to Production Files

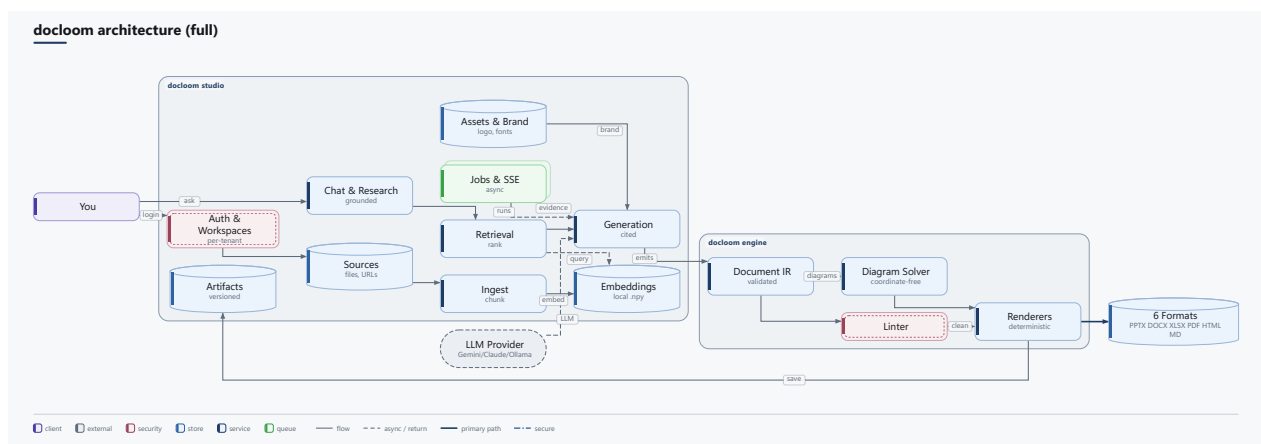
- LLM outputs are unstructured and unpredictable
- Manual formatting is error-prone and time-consuming
- Renderers need consistent, validated input
- Quality checks should happen before any file is written
- Users need native, editable formats—*not* rasterized images

Validated IR, Deterministic Renderers

- LLM emits schema-validated JSON (Pydantic IR)
- Deterministic linter checks layout, citations, contrast
- Findings loop back to model for self-correction
- Deterministic renderers: same IR = same output, always

The IR is deliberately non-recursive with plain tagged unions—*not* shaped to survive real LLM structured-output constraints.

docloom architecture, end to end



The full studio + engine architecture: auth, retrieval, jobs, generation, versioned artifacts, and deterministic rendering.

- Sources are chunked, embedded, and ranked, so only the most relevant evidence reaches the model
- The model emits validated IR, never files; the linter checks layout and citations before any render
- Deterministic renderers turn one IR into all six formats with no lossy conversion

Count What Matters: 14 Block Types, 6 Formats, 658+ Tests

14

Block Types

6

Output Formats

8

Slide Layouts

7

Diagram
Types Node

658+

Tests (Passing)

Generate Six Native Formats from One IR

- PPTX: native editable charts, autofit text, native shapes
- DOCX: hardened tables, repeating headers, fixed column widths
- XLSX: typed cells, formulas, number formats
- PDF: Typst compiled in-process, no LaTeX needed
- HTML: semantic markup with WCAG 2.x contrast
- Markdown: clean, readable prose

Turn Source Data into Grounded Artifacts

Add Sources

- Upload file, URL, or paste text
- Free web research agent included
- Hybrid BM25 + dense retrieval
- Automatic chunking at 1000 chars

Generate Artifacts

- Chat with grounded, cited answers
- 5 one-click guides (briefing, FAQ, timeline)
- 6 artifact kinds: deck, doc, sheet, diagram, infographic, podcast

Keep Text Legible at Any Scale

- Measured autofit: real text extents via FreeType against actual fonts
- WCAG 2.x contrast ratio checking: 4.5:1 for normal text
- Diagram legibility: labels never truncated, step down detail instead
- Table hardening: fixed widths, repeating headers, split prevention
- Geometric QA: off-slide detection, overlap checks, font sprawl

Design principle: lint before render. Only error severity blocks a render; warnings help improve.

Local-First, Offline-Ready, Multi-Tenant-Safe

- Runs fully offline against a local model (ollama, llama.cpp, etc.)
- SQLite default, Postgres for scale
- Session tokens stored only as SHA-256 hashes
- Cross-tenant access returns 404, not 403
- Secrets encrypted at rest with Fernet

One Command to Launch

Windows (PowerShell)

```
studio.ps1
```

macOS / Linux / Git Bash

```
studio.sh
```

- Installs dependencies, builds frontend, opens browser
- Default: `http://127.0.0.1:8899`
- Prerequisites: `uv`, Node 22+, `npm`

Set Up Providers, Storage, and Secrets

- Generation: `ollama`, `llama-server`, `lmstudio`, `openai`, `anthropic`, `gemini`
- Storage: `SQLite` (default) or `Postgres` via `DOCLOOM_DB_URL`
- Secrets: `Fernet` encryption, masked in API responses
- Optional extras: `[pdf,diagrams]` for engine; `[podcast,ingest]` for studio
- Settings resolve: per-user â†’ global â†’ built-in defaults

Design Principles: Built for Reliability

- Validated IR, not code: LLM output is data, never executable
- Deterministic renderers: same input = same output, always
- Lint before render: findings can loop back to model
- Nothing authored is lost: labels step down, never truncate
- Graceful degradation: optional deps improve output, never crash

Open Source, Local-First, Production-Ready

MIT License • Python 0.2.0 • 658+ Tests • Full-Stack SPA